Tetris 2.0

User Manual

A Tetris game with player accounts, hold and next queue, T-spins, combos, row bombs, music, themes, personal high scores, and optional cloud score sync.

Version: tetris_ver2.py
Players: one at a time, on one computer

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1. What You Need

- **Python 3** (a recent version, such as 3.10 or newer)
- **pygame**, the library that draws the game and plays sound
- A mouse and keyboard
- Sound is optional. If no audio device is found, the game stays silent and everything else still works.

Install pygame (once), in a terminal:

```
pip install pygame
```

On some computers the command is `pip3` or `python3 -m pip`.

2. Starting the Game

1. Put all the game files in one folder (see section 12).
2. Open a terminal in that folder.
3. Run:

```
python3 tetris_ver2.py
```

A window (700 x 660) opens on the **Sign In** page.

To close the game: click **Quit** on the Sign In page, or use the **Quit** buttons on the pause menu or game over screen.

Tip: if you want cloud score sync, set it up first (section 11).

3. Signing In and Creating an Account

The game always opens on the **Sign In** page. You use the mouse for almost everything.

Do this	To
Click a box	Type in it
Click **Show / Hide**	Reveal or hide your password
Click **Sign In**	Submit
Click **Create an account**	Open the Sign Up page
Click **Quit**	Close the game

Keyboard helpers: Tab and the **Up / Down arrows** move between boxes. **Enter** moves to the next box, and submits from the last box.

4. Creating an Account (Sign Up)

Click **Create an account**, then fill in:

- **Username:** 3 to 12 letters, numbers, or underscores. Not case sensitive (`Aiden` and `aiden` are the same player), but it keeps the capital letters you chose.
- **Password:** 6 to 64 characters. It can't be the same as your username.
- **Confirm password:** type the same password again.

Click **Sign Up**. You are signed in right away. **Back to Sign In** returns to the first page.

Error messages tell you what to fix, for example: *That username is taken*, *Passwords don't match*, *Password must be 6-64 characters*.

5. How Accounts Are Kept Safe (and Their Limits)

- Your password is **never stored**. The game keeps a random salt and a PBKDF2-HMAC-SHA256 hash in `users.json`.
- A wrong sign in always shows the same message: *Wrong username or password*. It never says which part was wrong.
- Accounts are **local to this computer**. They keep casual players out of each other's scores and settings.
- Anyone with access to the files can edit or delete them, so **do not reuse an important password here**.
- **There is no password recovery.** If you forget it, create a new account.

6. The Game Screen

Area	What it shows
Center	The 10 x 20 board, your falling piece, and a **ghost** outline showing where it will land
Top left: HOLD	The piece you saved with **C**
Right: NEXT	The next **3 pieces**, in order
Left, under HOLD	Score, Level, Lines, Pieces, Time, and Bombs (n / 3)
Bottom left	Reminders: pause, sound on/off, current theme, row bomb key
Bottom right	Who is signed in, and the **Sign Out** button
Top of the board	Pop-up text: SINGLE, TETRIS, T-SPIN, COMBO, BACK-TO-BACK, BOOM!, LEVEL UP!

7. Controls

Key	Action
Left / Right arrow	Move the piece
Up arrow	Rotate clockwise
Z	Rotate the other way
Down arrow (hold)	Soft drop: fall faster
Space	Hard drop: land instantly
C	Hold the piece
B	Use a row bomb
P or **Esc**	Pause / resume
M	Sound and music on / off
T	Switch theme (light, dark, retro)

8. Controls: Menus and Game Over

Key	When	Action
R	Paused, or game over	Restart
Q	Paused, or game over	Quit the game
L	Game over	Sign out

Mouse:

- Pause menu: click **Resume**, **Restart**, or **Quit**.
- Game over: click **Restart**, **Sign Out**, or **Quit**.
- During play: click **Sign Out** (bottom right) to return to the Sign In page.

Pausing hides the board, so you can't plan your next move while paused.

9. Pieces and the 7-Bag

There are 7 pieces: **I, O, T, S, Z, J, L**.

The game deals them with a **7-bag**: every bag contains each piece exactly once, in random order. So you never wait too long for a piece, and you never get long runs of the same one.

Pieces rotate with the standard **SRS system with wall kicks**: if a rotation would hit a wall or the stack, the game tries a few nearby positions so the turn still works. This is what makes tight spins possible.

10. Hold, Next Queue, and Ghost Piece

Hold (C): swap your falling piece into the hold box. If the box already has a piece, they trade places. You can hold **once per piece**; the next hold unlocks after the current piece locks.

Next queue: the right side always shows the next **3 pieces** in order, so you can plan ahead.

Ghost piece: the outline at the bottom of the board is exactly where your piece will land if you hard drop. Use it to line up drops precisely.

11. Lock Delay

A piece that lands **does not lock right away**. It stays movable for **half a second**, so you can slide or rotate it into place.

- Moving sideways or rotating on the ground **restarts the timer**.
- This can happen up to **15 times** per piece, so a piece can't stall forever.
- **Hard drop** (Space) locks instantly.

When the piece locks, completed rows flash, then disappear, and the next piece appears.

12. Scoring: Line Clears

All line clear points are **multiplied by your level**.

Lines cleared	Name	Base points
1	SINGLE	100
2	DOUBLE	300
3	TRIPLE	500
4	TETRIS	800

Dropping also scores: 1 point per row of soft drop, and 2 points per row of hard drop.

Example: a Tetris at level 3 scores 800 x 3 = 2,400.

13. Levels and Speed

- You gain a level every **10 lines**.
- A **LEVEL UP!** message appears and a short jingle plays.
- Pieces fall faster at every level, and the music gets faster too.

Level	Seconds per row (about)
1	1.00
5	0.36
10	0.06
Higher	never faster than 0.05

The game over screen reports the level you reached.

14. T-Spins

Rotate a **T piece** into a tight slot so that **3 of the 4 corners** around it are blocked (walls and the floor count). The last thing you did must be the rotation.

Result	No lines	1 line	2 lines	3 lines
T-SPIN	400	800	1,200	1,600
T-SPIN MINI	100	200	400	

Points are multiplied by level. A **mini** T-spin is one where the two corners on the pointed side are not both blocked. Moving the piece after rotating cancels the T-spin.

15. Combos and Back-to-Back

Combo: clear lines with **consecutive pieces** and each clear after the first adds bonus points: **50 x combo step x level**. A piece that clears nothing ends the combo. The screen shows *COMBO x2*, *COMBO x3*, and so on.

Back-to-back (B2B): a **Tetris** or a **T-spin line clear** is a "difficult" clear. Do one right after another difficult clear and it scores **1.5x**. The screen shows *BACK-TO-BACK*.

- A normal single, double, or triple **ends** the chain.
- A piece that clears nothing does **not** end it.

16. Row Bombs

Earn a bomb: every **Tetris** or **T-spin line clear** gives you one row bomb. You can hold up to **3**. The HUD shows *Bombs: n/3*, and a chime plays when you earn one.

Use a bomb (B): the **bottom 2 rows** are blasted away and everything above drops down. The screen shows *BOOM!*

- You score **50 points per blasted row that had blocks, times your level.**
- A bomb is **never wasted**: it won't fire on an empty floor, and you keep it.
- Bombs don't work while paused or after game over.

Best use: rescue a stack that is getting too high, or open up a messy bottom.

17. Sound and Music

Sound effects (generated in code, no audio files): moving, rotating, holding, locking, clears, T-spins, Tetris, level up, bomb earned, bomb blast, and game over.

Background music is a short loop that plays while you play:

- It gets **faster with each level**, up to level 10.
- It **pauses** when you pause and **stops** at game over.

Press M to turn all sound and music on or off. The setting is saved for your account.

No audio device? The game stays silent and runs normally.

18. Themes

Press **T** to cycle through three looks:

Theme	Look
Light	The default: light background, bright piece colors
Dark	Dark background with soft contrast
Retro	Game Boy style greens

Your theme and sound setting are saved **per player** in `settings/<username>.json` and come back the next time you sign in. The Sign In pages always use the light theme.

19. Pause Menu

Press **P** or **Esc** to pause. The board is hidden and a menu appears:

Button	Key	What it does
Resume	P or Esc	Continue playing
Restart	R	Save your score and start a new game
Quit	Q	Save your score and close the game

While paused, the time and gravity stop, and the music pauses.

Important: closing the window with the X button does **not** save an unfinished run. Use **Pause > Quit** or **Sign Out** to keep your score.

20. Game Over

The game ends when a new piece has no room to appear.

The game over screen shows:

- **Pieces** placed, **Lines** cleared, **Level** reached, and **Time** played
- Your **top 5 scores** (your new score is highlighted in red if it made the list)
- Buttons: **Restart** (R), **Sign Out** (L), **Quit** (Q)

Your score is saved automatically. A score of 0 is not saved.

21. Signing Out

Click **Sign Out** (bottom right) at any time, or press **L** after game over.

- Your **current score is saved first**, so you don't lose the run.
- You return to the Sign In page, where another player can sign in.
- Each player has their own scores and settings.

22. High Scores

Each player's **top 5 scores** are saved in `scores/<username>.json` next to the game. They appear on the game over screen under **YOUR HIGH SCORES**.

- Scores are kept separately for every account.
- Only your best 5 are kept.
- With cloud sync turned on, the title changes to **HIGH SCORES (CLOUD)** after a successful sync.

23. High Score Cloud Sync: What It Does

Cloud sync lets you keep the same top scores on **more than one computer**. It uses a small score server that comes with the game.

- It is **optional**. If you don't set it up, nothing changes.
- Scores are merged when you **sign out, quit, or reach game over**.
- The same score is **never counted twice**, even if you sync many times.
- If the server is off or unreachable, the game simply keeps your local scores. Syncing can pause the game for up to about 1.5 seconds.

The server is one you run yourself. It is not a public online service.

24. Cloud Sync: Setup

Step 1: start the server on a computer the players can reach:

```
python3 cloud_server.py --port 8765 --key mysecret
```

Step 2: tell the game where it is, before launching it (macOS / Linux):

```
export TETRIS_CLOUD_URL=http://127.0.0.1:8765
export TETRIS_CLOUD_KEY=mysecret
python3 tetris_ver2.py
```

On Windows PowerShell use `$env:TETRIS_CLOUD_URL="http://127.0.0.1:8765"` and `$env:TETRIS_CLOUD_KEY="mysecret"`.

25. Cloud Sync: Safety Notes

- By default the server listens on **127.0.0.1**, so only this computer can reach it.
- To allow other computers, add `--host 0.0.0.0`, and **always use a key**.
- The key is a shared password. If the server is not on your own network, put it behind **https**.
- Server data is stored in `cloud_scores.json`. Add it to `.gitignore`.
- The server checks every request: usernames, score values, and request size are limited, and bad requests are refused.

26. Cloud Sync: The API (Reference)

JSON in, JSON out. Everything except `/health` needs the `X-API-Key` header when the server has a key.

Request	What it does
`GET /health`	Returns `{"status": "ok"}`
`GET /scores/<user>`	The player's top scores
`POST /scores/<user>` with `{"score": 1200}`	Adds one score
`POST /sync/<user>` with `{"scores": [900, 300]}`	Merges a list, returns the top 5

Replies: `200` success, `400` bad request, `401` wrong or missing key, `404` unknown path or bad username.

27. Files the Game Creates

File or folder	What it holds
`users.json`	Accounts: usernames, salts, password hashes
`scores/<username>.json`	That player's top 5 scores
`settings/<username>.json`	That player's theme and sound setting
`cloud_scores.json`	Server only: all players' cloud scores

These are created automatically next to the game. Don't commit them to git (add to `.gitignore`). To reset a player, delete their file in `scores` or `settings`.

28. Troubleshooting: Starting the Game

Problem	Fix
`No module named pygame`	Run `pip install pygame`
`tetris_ver2.py` not found	Open the terminal **in the game folder**
Window doesn't open	Make sure Python and pygame installed in the same environment
`No module named cloud_sync`	Keep `cloud_sync.py` in the same folder as `tetris_ver2.py`
Slow or choppy	Close other programs; the game targets 60 frames per second

29. Troubleshooting: Accounts and Scores

Problem	Fix
"Wrong username or password"	Check spelling; the username is not case sensitive, the password is
"That username is taken"	Pick another name; names are not case sensitive
Forgot a password	There is no recovery; create a new account
My score didn't save	Use Pause > Quit or Sign Out. Closing the window with X loses an unfinished run
Scores missing on another computer	Accounts and scores are local unless cloud sync is on
Score of 0 not shown	Zero scores are never saved

30. Troubleshooting: Sound and Cloud Sync

Problem	Fix
No sound or music	Press **M** (it may be off). Check your computer's audio device
Music cuts out when paused	Normal: it pauses with the game
Cloud title never appears	The sync failed. Check the server is running, the URL and key match, and the firewall allows the port
Short freeze at game over	The game waited for the server (up to about 1.5 seconds)
`401` errors	The key in `TETRIS_CLOUD_KEY` doesn't match the server's `--key`

Quick check: open `http://127.0.0.1:8765/health` in a browser. It should show `{"status": "ok"}`.

31. FAQ

Can two people play at once? No. Players take turns by signing in and out.

Does my score change if I pause? No. Time, gravity, and music all stop.

How do I get more bombs? Tetrises and T-spin line clears. Holding is capped at 3.

Why can't I hold twice in a row? One hold per piece is the standard rule; it unlocks when the piece locks.

Is cloud sync required? No, it's optional and off by default.

Is 2-player battle mode available? Not yet. Background music and bombs are done; battle mode is still planned.

32. For Developers: Tests

Run all tests from the game folder:

```
python3 -m unittest test_tetris_render test_tetris_auth \
  test_tetris_features test_tetris_cloud test_tetris_extras
```

Test file	Covers
`test_tetris_render`	Drawing and the score file
`test_tetris_auth`	Accounts, passwords, sign in pages
`test_tetris_features`	Pause, T-spins, combos, lock delay, themes, sound
`test_tetris_cloud`	Score merging, the server, the sync client
`test_tetris_extras`	Next queue, row bombs, music

33. For Developers: Project Files

File	Purpose
`tetris_ver2.py`	The game: board, pieces, scoring, drawing, accounts, play loop
`cloud_sync.py`	Score client and merge rules (no pygame)
`cloud_server.py`	The score server (standard library only)
`README.md`	Short reference
`test_*.py`	Automated tests

Key settings are constants at the top of `tetris_ver2.py` (window size, lock delay, scoring tables, bomb limits, music tempo), so you can tune the game without changing the logic.

Quick Reference Card

Move	Left / Right
Rotate	Up (clockwise), Z (other way)
Drop	Down (soft), Space (hard)
Hold	C
Row bomb	B
Pause	P or Esc
Sound / music	M
Theme	T
Restart / Quit / Sign out	R / Q / L (when paused or game over)

Goal: clear lines, chain combos, earn bombs, and beat your high score.